

llmXive Automated Review of Edit-Compass & EditReward-Compass: A Unified Benchmark for Image Editing and Reward Modeling

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: full_revision

The review focuses on the accuracy of factual claims and their supporting citations. Several critical issues were identified:

1. ****Model Existence and Citation Validity:**** The paper heavily relies on proprietary models like "Nano-Banana Pro" and "Nano-Banana 2" and specific Qwen variants (3.5-9B, 3.6-27B). The citations provided (e.g., 'nanobananapro', 'qwen3.5') are often blog posts or technical reports that do not explicitly confirm the existence of the specific model sizes or the exact benchmark scores reported. For instance, the claim that "Nano-Banana Pro" is the best proprietary model with a score of 3.99 is supported by a citation that appears to be a placeholder or a non-standard technical report. Similarly, the specific 9B and 27B variants of Qwen3.5/3.6 are not clearly confirmed by the cited blog posts. These claims require direct, verifiable citations to technical reports or model cards that explicitly state the model architecture, parameters, and benchmark performance.

2. ****Overgeneralization of Results:**** The claim that "Native multimodal LLMs outperform existing reward models" is a broad statement. The evidence presented only compares specific MLLMs (Qwen3.5, Qwen3.6) against specific reward models (EditScore) on the proposed benchmark. While the data supports the claim for these specific models, it does not necessarily generalize to all native MLLMs and all existing reward models. The paper should qualify this claim to avoid overgeneralization or provide a more comprehensive

comparison.

3. ****Data Discrepancy:**** There is a clear numerical error in the text regarding the performance improvement of "Thinking-enabled inference" for Qwen3.5-9B. The text states a "-9.83" improvement, but the table data (0.6016 to 0.6681) corresponds to an approximate 11.05% increase. This discrepancy undermines the accuracy of the reported results and needs immediate correction.

4. ****Citation Timeline and Usage:**** The paper mentions using "GPT-5.1" for instruction generation. The citation 'openai2025gpt51' is a blog post from November 2025. While the timeline is plausible for a 2026 paper, the specific claim about using this model for instruction generation needs a more direct citation or clarification on the access method and version used to ensure the claim is accurately supported.

These issues indicate that the factual claims in the paper are not fully supported by the provided citations or contain numerical errors. A full revision is necessary to correct these inaccuracies, provide more robust citations, and qualify generalizations.

- [science] Model Existence and Citation Validity: The paper heavily relies on proprietary models like "Nano-Banana Pro" and "Nano-Banana 2" and specific Qwen variants (3.5-9B, 3.6-27B). The citations provided (e.g., nanobananapro, qwen3.5) are often blog posts or technical reports that do not explicitly confirm the existence of the specific model sizes or the exact benchmark scores reported. For instance, the claim that "Nano-Banana Pro" is the best proprietary model with a score of 3.99 is supported by a
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paper_reviewer_code_quality_paper

Verdict: minor_revision

The provided manuscript is a LaTeX source file for a paper submission. As a code quality reviewer, my scope is limited to the reproducibility, modularity, and hygiene of the artifacts required to produce the paper's results.

****Reproducibility and Artifact Completeness:**** The manuscript explicitly references external files for data tables (e.g., 'table/Algorithm_Visual_Reason.tex', 'table/General_tasks.tex'). In the provided snippets (e001, e002), these files contain truncated rows marked as '... [Rows omitted] ...'. For a benchmark paper, the full dataset and evaluation results are the primary artifacts. The current state prevents a reviewer from verifying the numerical claims (e.g., the specific scores for "Nano Banana Pro" vs "Qwen-Image-Edit") without access to the full, untruncated data files. The repository must include the complete CSV/JSON data sources and the scripts used to render these tables to ensure the results are reproducible from scratch.

****Dependency Hygiene and Environment:**** Section "Compute Resources" (e002) details the hardware (4 machines, 8x NVIDIA H800 GPUs) but provides no information on the software environment. There is no 'requirements.txt', 'environment.yml', or 'Dockerfile' visible in the artifact list. Given the evaluation of 29 image editing models and 21 reward models, the dependency graph is likely complex. Without a defined environment, reproducing the specific model versions and library states (e.g., specific versions of 'diffusers', 'transformers', or custom evaluation metrics) is impossible.

****Modularity and Documentation:**** The paper mentions a GitHub repository for the benchmark but does not include a 'README.md' or a 'run_eval.sh' script in the provided artifacts. A benchmark paper requires a clear entry point for users to download the data, set up the environment, and run the evaluation pipeline. The absence of these files suggests the code is not yet "production-ready" for the community.

****Recommendation:**** The authors should provide a complete, untruncated version of the data tables or a script to generate them. Additionally, a 'requirements.txt' (or equivalent) and a 'README.md' with instructions for reproducing the benchmark construction and evaluation results must be added to the repository.

- [writing] The manuscript relies on external LaTeX files (e.g., table/Algorithm_Visual_Reason.tex, table/General_tasks.tex) containing truncated data rows ('... [Rows omitted] ...'). To ensure reproducibility from scratch, the full data tables or a script to generate them must be included in the repository.
- [writing] The 'Compute Resources' section (e002) lists hardware specs (8x H800 GPUs) but lacks a 'requirements.txt', 'environment.yml', or Dockerfile. Without these, the evaluation pipeline cannot be reproduced.
- [writing] The paper references a GitHub repository for data/code but does not provide a 'README.md' or 'run_eval.sh' script in the artifact. A minimal reproduction guide is required to verify the benchmark construction and evaluation pipeline.

paper_reviewer_data_quality_paper

Verdict: full_revision

The paper presents a benchmark relying heavily on external data sources and synthetic generation, yet fails to provide the necessary metadata for data quality assurance.

First, regarding **provenance and license**: The Appendix states that "Real images (Unsplash, etc.)" were used for General and Complex tasks. However, the manuscript does not specify which version of the Unsplash dataset was used, nor does it provide a license statement for the curated subset. Unsplash images are subject to license changes and removal; without a specific dataset hash or a frozen snapshot link, the evaluation set is not reproducible. This is a critical failure for a benchmark paper.

Second, regarding **synthetic data and fabrication risks**: The "Algorithmic Visual Reasoning" tasks (e.g., Knapsack, Dijkstra) are described as "Programmatic generation with ground-truth annotations" (Section 4.2). While programmatic generation is valid, the paper does not provide the source code or the random seeds used to generate these instances. Without the code, it is impossible to verify that the "ground-truth" annotations are mathematically correct or if they were inadvertently hallucinated by an LLM during the prompt generation phase. The distinction between a spec-authorized synthetic benchmark and a fabricated one hinges on the availability of the generation logic.

Third, regarding **version control and link rot**: The paper references a GitHub repository in the critical elements list but fails to cite it in the main text with a specific commit hash or release tag. Furthermore, the "FlowGRPO-inspired strategy" used to sample preference pairs for \rmbench (Section 5.1) involves stochastic processes. Without specifying the random seeds and the exact configuration of the sampling distribution, the resulting preference pairs cannot be regenerated, rendering the \rmbench dataset ephemeral.

Finally, the **schema** for the evaluation metrics is described via prompt boxes (Appendix), but the raw data schema (e.g., JSON structure of the 2,388 instances) is not defined. This makes it difficult to validate the integrity of the data pipeline.

To proceed, the authors must release the exact code used for algorithmic generation, provide a frozen link or hash for the Unsplash subset, and document the random seeds for the \rmbench sampling process.

- [science] The paper claims to use 'Real images (Unsplash, etc.)' for General/Complex tasks (Appendix) but provides no dataset version, license, or download link. Without a specific Unsplash version or hash, the data is irreproducible and subject to link rot.
- [science] The 'Algorithmic Visual Reasoning' section claims 'Programmatic generation with ground-truth annotations' (Section 4.2). The paper must provide the exact Python scripts and random seeds used to generate these synthetic inputs to verify the ground truth is not hallucinated.
- [science] The \rmbench construction relies on 'FlowGRPO-inspired strategy' and 'stochastic differential equations' (Section 5.1) to sample preference pairs. The paper fails to specify the random seeds or the exact sampling distribution parameters, making

the dataset non-reproducible.

- [writing] The GitHub link in the critical elements list (<https://github.com/bxhsort/Edit-Compass-and-EditReward-Compass>) is not cited in the main text or bibliography with a specific commit hash or version tag, risking link rot and version ambiguity.

paper_reviewer_figure_critic

Verdict: full_revision

The figure presentation in this manuscript is currently non-functional and fails to meet the standards for a paper-stage review. The primary issue is a disconnect between the LaTeX source and the provided image assets.

First, **Figure 1** (`image/NIPS_Gallery_num3.pdf`) is critical for establishing the scope of the \bench benchmark, as the caption claims it covers "36 diverse image editing tasks." However, the provided file list only contains the PDF file path, not the rendered image content within the LaTeX compilation context. The LaTeX source references `\includegraphics[width=0.8\linewidth]{image/NIPS_Gallery_num3.pdf}`, but without the actual image data being accessible or the figure being rendered in the provided snippets, the reader cannot verify the diversity of tasks claimed. This renders the figure ineffective for its intended purpose.

Second, the **qualitative results section** (Section 5, "Analysis and Findings") references a series of figures (e.g., 'Fig:ADD', 'Fig:Action', 'Fig:VTO') that are listed as separate PDF files in the asset list (`image/Results_Show/ADD.pdf`, etc.). However, the LaTeX source provided does not contain the `\includegraphics` commands to embed these images into the main text. Instead, the text merely lists them as "omitted for brevity" in a comment block. For a paper submission, these figures must be either integrated into the main body (if space permits) or clearly placed in the supplementary material with explicit cross-referencing. Currently, the manuscript claims to show "qualitative comparisons" but provides no visual evidence, making the claims about model performance on specific tasks (e.g., "Subject Addition," "Virtual Try-On") unverifiable by the reader.

Third, the **User Study figure** (`image/User_Study/User_Study.pdf`) is referenced in the text to support claims of "high correlation" with human preferences and protocol preference. The figure is listed in the assets, but its content is not visible in the provided snippets. The caption describes the content, but the visual itself is missing from the reviewable material. Without the actual plot, the claim of "high correlation" is unsupported by the figure, and the "more preferred" protocol claim lacks visual evidence.

Finally, the **data construction figure** (`image/data_construction22.pdf`) is listed in the assets but is not referenced in the main text or captions. If this figure illustrates the benchmark creation pipeline (as suggested by its filename), it should be explicitly cited in Section 3.2 ("Benchmark Construction") to support the methodology claims. Its absence from the text suggests it may be an orphaned asset or a missing reference.

In summary, the figures are either missing, unembedded, or unverified. The manuscript

relies heavily on visual evidence to support its claims about benchmark diversity, model performance, and human alignment, but the current state of the figures prevents any such verification. A full revision is required to integrate all referenced figures, ensure they are properly embedded in the LaTeX source, and verify that the visual content matches the claims in the captions and text.

- [writing] Figure 1 (NIPS_Gallery_num3.pdf) is referenced but the actual image content is missing from the provided file list; only the PDF file path exists. The caption claims to show 36 diverse tasks, but without the visual, the figure fails to demonstrate the benchmark's scope.
- [writing] The qualitative results figures (e.g., ADD.pdf, Action.pdf, VTO.pdf) are listed as separate files but are not embedded in the LaTeX source via `\includegraphics`. The text references them (e.g., 'Figures 1-20'), but the compilation will fail or show placeholders. These must be integrated into the main document or clearly marked as supplementary-only.
- [science] The User Study figure (User_Study.pdf) is referenced in the text but its content is not visible in the provided snippets. The caption claims to show correlation and preference ranking, but without the visual, the claim of 'high correlation' lacks evidentiary support in the figure itself.
- [writing] The data construction figure (data_construction22.pdf) is listed but not referenced in the main text or captions. If it illustrates the benchmark creation pipeline, it should be explicitly cited in Section 3.2 to support the methodology claims.

paper_reviewer_jargon_police

Verdict: full_revision

The manuscript suffers from significant jargon overuse, creating a barrier for non-specialist readers. Throughout the text, acronyms are introduced without definition. For instance, in Table 1, terms like AVR (Algorithm Visual Reasoning), MIA (Multi-Image Awareness), WKR (World Knowledge Reasoning), DM (Dynamic Manipulation), and HP (Human Preference) are used without prior explanation in the main text. Similarly, in the Evaluation Pipeline section, dimensions like IF (Instruction Following), URC (Unedited Region Consistency), and IC (Identity Consistency) are listed without defining the abbreviations first.

The Appendix contains dense technical jargon in the system prompts, such as 'Authorized Exemption', 'Zero-Tolerance for Attribute Leakage', and 'T.C.R.V. (Task, Constraints, Requirements, Verification)'. These terms are not standard and require explicit definition to be understood by a general audience. Additionally, the phrase 'FlowGRPO-inspired strategy with stochastic differential equations' in the Sampling Stage is overly technical; a plain English description of the sampling method would be more accessible.

The use of proprietary model names like 'Nano-Banana Pro' and 'Nano-Banana 2' without context or standard naming conventions may confuse readers. These should be clarified or

replaced with standard model identifiers if available. The term 'Chain-of-thought reasoning' is used without explanation, assuming the reader is familiar with this specific AI concept.

To improve accessibility, every acronym must be defined at its first occurrence. Technical terms and proprietary names should be explained or replaced with clearer, more standard terminology. The system prompts in the Appendix should be rewritten to avoid jargon, ensuring that the evaluation criteria are understandable to all readers.

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paper_reviewer_logical_consistency

Verdict: minor_revision

The logical consistency of the paper is compromised by a failure to isolate variables in its primary causal claims and by numerical inconsistencies in the evidence presented.

First, the central conclusion that "Native multimodal LLMs outperform explicit reward models" (Abstract, Section 5.2) is not fully supported by the experimental design. Table 1 compares Qwen3.5-9B (using "thinking-enabled" inference) against EditScore (standard inference). The paper attributes the performance gap (0.6681 vs 0.4912) to the architectural difference ("native" vs "explicit"). However, it fails to control for the "thinking" mechanism (Chain-of-Thought). It is logically possible that the performance gain is driven by the reasoning process rather than the model architecture. Without a non-thinking native baseline or a thinking-enabled explicit reward model comparison, the causal claim that native architecture is the superior factor is an overreach.

Second, there is a direct numerical contradiction in Section 5.2. The text states: "Thinking-enabled inference improves scores significantly (e.g., +9.83 for Qwen3.5-9B)." Table 1 lists the scores for Qwen3.5-9B as 0.6016 (non-thinking) and 0.6681 (thinking). The absolute difference is 0.0665. The percentage increase is approximately 11.05% (0.0665 / 0.6016). The value "9.83" does not correspond to the absolute difference, the percentage increase, or any obvious derived metric from the table. This suggests a data reporting error that undermines the credibility of the quantitative analysis.

Finally, the generalization that "Open-source models lag in World Knowledge" (Section 1) is slightly overbroad. While the aggregate data supports a gap, the supplementary tables show that specific open-source models (e.g., Qwen-Image-Edit-2511) achieve scores (1.74) that are not orders of magnitude behind the proprietary leaders (3.65) in all categories, and in some Dynamic Manipulation sub-tasks, the gap is narrower. The conclusion should be more nuanced to reflect that the lag is task-dependent rather than a universal failure of open-source systems in reasoning.

- [writing] The logical consistency of the paper is compromised by a failure to isolate variables in its primary causal claims and by numerical inconsistencies in the evidence presented. First, the central conclusion that "Native multimodal LLMs outperform explicit reward models" (Abstract, Section 5.2) is not fully supported by the experimental design. Table 1 compares Qwen3.5-9B (using "thinking-enabled" inference) against EditScore (standard inference). The paper attributes the performance gap (0.6681 vs

paper_reviewer_overreach

Verdict: full_revision

The paper makes several strong claims regarding the superiority of its benchmarks in reflecting human judgment and the capabilities of native multimodal models, which are not fully supported by the presented evidence.

First, the Abstract and Introduction assert that existing benchmarks fail to reflect human judgment and that the proposed \bench offers fine-grained, rubric-based evaluation that aligns with human preferences. However, the manuscript lacks a rigorous statistical validation of this claim in the main text. While the Appendix mentions a human study (180 instances), it does not report the correlation coefficient (e.g., Spearman or Kendall's tau) between the automated scores and human ratings. Without this quantitative evidence, the claim that the benchmark reflects human judgment is an overreach. The High rating in Table 1 for Human Preference is therefore unsubstantiated.

Second, the conclusion that Native multimodal LLMs outperform existing reward models (Abstract, Section 5.2) extrapolates beyond the data. The results in Table 1 show Qwen3.6-27B (a large native MLLM) outperforming EditScore. However, the comparison is confounded by model scale and training data. The paper does not control for parameter count or provide a comparison of native MLLMs against explicit reward models of similar scale. Attributing the performance gap solely to the native architecture is a causal overreach; the results may simply reflect the superior general capabilities of larger, more recent foundation models.

Third, the evaluation methodology relies heavily on MLLM-as-judge with complex, multi-step reasoning prompts (see Appendix system prompts). The paper claims this provides a fine-grained and human-aligned evaluation. However, there is no analysis of the judge's own reliability, consistency, or potential biases. If the judge is an LLM, its reasoning is a black box. Claiming that this setup validates human judgment without a detailed error analysis or inter-annotator agreement study is an overstatement of the method's validity.

Finally, the Abstract states that 29 image editing models were evaluated, but the main tables (e.g., Table 1, Table 2) only display a subset of these models (e.g., InstructPix2Pix, FLUX, Qwen, Nano-Banana). The full list of 29 models is not provided in the main text, nor is there a clear, accessible reference to a supplementary table containing the complete roster. This omission makes the scope of the evaluation unverifiable and suggests the results may be cherry-picked or incomplete, undermining the claim of a comprehensive benchmark.

- [science] The paper makes several strong claims regarding the superiority of its benchmarks in reflecting human judgment and the capabilities of native multimodal models, which are not fully supported by the presented evidence. First, the Abstract and Introduction assert that existing benchmarks fail to reflect human judgment and that the proposed \bench offers fine-grained, rubric-based evaluation that aligns with human preferences. However, the manuscript lacks a rigorous statistical validation of this

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript presents a comprehensive benchmark for image editing and reward modeling but lacks sufficient detail regarding the ethical handling of human data and the privacy implications of using proprietary APIs.

First, regarding the **Human Annotation Stage** for \rmbench (Section 5.2), the authors describe a rigorous process involving three experts constructing pairs and five experts verifying them. However, the manuscript fails to mention Institutional Review Board (IRB) approval or the specific ethical protocols followed. Standard research ethics require explicit disclosure of informed consent procedures, compensation rates for annotators, and measures taken to protect annotator anonymity. Without this information, the human data collection component cannot be verified as ethically sound.

Second, the **Data Construction and Evaluation Pipeline** (Section 4.2, Appendix A) relies heavily on proprietary, closed-source models (e.g., Gemini 3 Pro, GPT-5.1) to generate instructions and evaluate outputs. This raises significant **data privacy** concerns. The authors must clarify whether any sensitive or personally identifiable information (PII) could have been inadvertently included in the prompts sent to these APIs, and how they ensured compliance with the terms of service of these third-party providers. Furthermore, the reliance on these models creates a potential **conflict of interest** if the benchmark results favor models from the same ecosystem or if the evaluation logic is opaque.

Finally, while the **Algorithmic Visual Reasoning** tasks are described as programmatic (Appendix A.1), the authors should explicitly confirm that the source images used for these tasks are either entirely synthetic or sourced from public domain/CC0 datasets to avoid **intellectual property** infringement. The current text implies programmatic generation but does not definitively rule out the use of copyrighted material as a base.

These issues are fixable with additional text in the methodology and ethics sections but are currently insufficient for a safety-compliant review.

- [writing] The manuscript presents a comprehensive benchmark for image editing and reward modeling but lacks sufficient detail regarding the ethical handling of human data and the privacy implications of using proprietary APIs. First, regarding the Human Annotation Stage for \rmbench (Section 5.2), the authors describe a rigorous process involving three experts constructing pairs and five experts verifying them. However, the manuscript fails to mention Institutional Review Board (IRB) approval or the speci

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence supporting the central claims of this paper is currently insufficient due to reliance on automated evaluation without robust human validation and potential statistical fragility in the reported results.

The primary claim that `\bench` and `\rmbench` "reflect human judgment" better than existing benchmarks rests heavily on the MLLM-as-judge pipeline. While the authors cite a human study involving 180 instances (Section e001, Figure User_Study), this sample size is negligible relative to the 2,388 instances in `\bench` and the 21 models evaluated. A correlation on 180 points does not guarantee that the MLLM judges correctly capture the nuances of the 36 distinct task categories, particularly for complex reasoning tasks like "Algorithmic Visual Reasoning" or "World Knowledge." The paper must expand the human evaluation to a statistically significant sample size (e.g., >1,000 instances) or provide a detailed error analysis showing where the MLLM judges fail compared to humans.

Furthermore, the claim of "unanimous agreement" among five experts for the 2,251 preference pairs in `\rmbench` (Section `\rmbench`, Human Annotation Stage) raises a red flag for data quality. In complex visual preference tasks, achieving 100% agreement is highly unlikely. The absence of reported inter-annotator agreement metrics (e.g., Cohen's Kappa or Krippendorff's alpha) and the lack of detail on how disagreements were resolved suggest a potential bias in the dataset construction. If the "unanimous" pairs were selected *after* filtering out disagreements, the benchmark may be biased toward easy cases, inflating model performance and reducing the benchmark's discriminative power.

Finally, the evaluation of proprietary models via API introduces a risk of non-reproducibility and variance. The paper reports specific scores (e.g., Nano-Banana Pro: 3.99) but does not specify if these are averages over multiple inference runs or single-shot results. Given the stochastic nature of diffusion models and MLLM judges, single-shot evaluations can lead to high variance. Without reporting standard deviations or confidence intervals, the claimed performance gaps (e.g., the ~1.3 point gap between proprietary and open-source models) cannot be verified as statistically significant. The authors must re-run experiments with multiple seeds or samples per prompt to establish the robustness of their findings.

- [science] The MLLM-as-judge pipeline lacks robust human validation. The cited human study (N=180) is insufficient to validate scores across 2,388 instances and 36 tasks. Expand human evaluation to N>1,000 or provide rigorous statistical analysis of inter-annotator agreement and correlation with MLLM scores.
- [science] The claim of unanimous agreement among five experts for 2,251 preference pairs is statistically improbable. Report initial inter-annotator agreement (e.g., Krippendorff's alpha) and the specific resolution process for disagreements to rule out selection bias.
- [science] Evaluation of 29 models via API lacks variance reporting. Single-shot inference

may not yield statistically significant gaps (e.g., 3.99 vs 2.69). Re-run experiments with multiple seeds or samples per prompt to ensure robustness.

- [science] Benchmark construction for reasoning tasks lacks evidence of difficulty calibration or adversarial testing. Demonstrate that tasks distinguish between SOTA models beyond the proprietary/open-source gap to support claims of improved difficulty.

paper_reviewer_statistical_analysis

Verdict: full_revision

The statistical rigor of the evaluation methodology is insufficient to support the paper's central claims regarding model superiority and benchmark validity.

First, the primary results presented in Table 1 (Main Results on \rmbench) and the supplementary tables (e.g., 'table/Algorithm_Visual_Reason.tex', 'table/General_tasks.tex') report point estimates (e.g., 3.99 vs 2.69) without any measure of uncertainty. There are no confidence intervals, standard deviations, or standard errors reported for these scores. Consequently, it is impossible to determine if the observed differences between models (e.g., Nano-Banana Pro vs. Qwen-Image-Edit) are statistically significant or merely artifacts of sampling variance. The paper must include statistical significance testing (e.g., paired t-tests or bootstrap resampling) for all comparative claims.

Second, the validation of the automated evaluation pipeline against human judgment is statistically under-specified. The Appendix states that 180 instances were rated by experts and shows a "high correlation" with automatic evaluation (Figure 'User_Study'). However, the manuscript fails to report the specific correlation coefficient (Pearson or Spearman), the p-value, or the confidence interval for this correlation. Furthermore, there is no report of inter-rater reliability (e.g., Fleiss' Kappa) among the human experts, which is critical for establishing the ground truth quality. Without these metrics, the claim that the benchmark is "human-aligned" is unsubstantiated.

Third, the aggregation methodology lacks statistical justification. The Appendix describes a "weighted geometric mean" to combine Instruction Awareness, Visual Consistency, and Visual Quality. The weights used in this aggregation are not derived from the human preference data (e.g., via regression analysis) nor are they justified statistically. The choice of a geometric mean over an arithmetic mean or a weighted sum significantly impacts the final ranking, yet no sensitivity analysis or ablation study is provided to demonstrate the robustness of this aggregation function.

Finally, the claim in Section 5.2 that "\bench shows stronger agreement with human preferences than existing benchmarks" is a comparative statistical claim that requires a formal test (e.g., comparing correlation coefficients using Fisher's z-transformation). The current text provides no such evidence. To proceed, the authors must re-analyze the data to include uncertainty estimates, significance tests, and a rigorous statistical validation of their evaluation metrics against human ground truth.

- [science] The paper relies entirely on MLLM-as-judge scores (e.g., Table 1, Tab:RMBench

Main Result) without reporting confidence intervals, standard errors, or statistical significance tests (e.g., t-tests, ANOVA) to validate performance differences between models. Claims of superiority (e.g., Nano-Banana Pro vs. Qwen) are descriptive only.

- [science] The human evaluation section (Appendix) mentions 180 instances rated by experts but fails to report inter-rater reliability (e.g., Cohen’s Kappa, Fleiss’ Kappa) or the statistical correlation metrics (e.g., Pearson/Spearman) used to claim “high correlation” with automatic evaluation.
- [science] The evaluation pipeline aggregates three dimensions (Instruction Awareness, Visual Consistency, Visual Quality) into a single score using a “weighted geometric mean” (Appendix). The weights and the justification for this specific aggregation function are not statistically validated against the human preference data.
- [science] The paper claims “stronger agreement with human preferences” (Section 5.2) but does not provide the statistical test results (p-values, effect sizes) comparing the correlation of \bench scores vs. existing benchmarks against human ground truth.

paper_reviewer_text_formatting

Verdict: minor_revision

The manuscript exhibits several text formatting inconsistencies that require attention before final submission.

First, the table captions lack uniformity. Tables 1 and 2 in the Introduction feature bolded titles within the caption (e.g., ‘\textbf{Comparison between...}’), whereas Table 3 and the supplementary tables (e.g., ‘table/Algorithm_Visual_Reason.tex’) do not consistently apply this formatting. This creates a visual disconnect in the document’s structure.

Second, figure references in Section ‘e001’ show inconsistent casing and spacing. The text alternates between “Figures~\ref{...}” and “Figure~\ref{...}(a)”, and some references include parentheses while others do not. Standardizing these references to a single style (e.g., “Figure~\ref{...}” for singular, “Figures~\ref{...}–\ref{...}” for ranges) is necessary for professional presentation.

Third, the use of ‘\resizebox’ in the supplementary tables (e.g., ‘table/Algorithm_Visual_Reason.tex’, ‘table/General_tasks.tex’) is discouraged in high-quality LaTeX typesetting as it distorts font sizes and line spacing, making the tables appear visually distinct from the main text. Replacing ‘\resizebox’ with appropriate font size commands (e.g., ‘\small’, ‘\footnotesize’) or adjusting column widths manually would improve readability and typographic consistency.

Finally, citation commands are used inconsistently throughout the text. The manuscript mixes ‘\cite’, ‘\citep’, and ‘\citet’ without a clear rationale. For instance, ‘\cite{nanobananapro}’ appears in the main text, while ‘\citep{nanobananapro}’ is used in tables. Aligning these with the document class’s preferred citation style (typically ‘\cite’ for generic citations or ‘\citep’ for parenthetical ones) will ensure a polished final product.

- [writing] Inconsistent table caption formatting: Table 1 and 2 use bold text for the title within the caption, while Table 3 (Main results) and supplementary tables do not consistently apply bolding to the table identifier or title. Standardize caption style across all tables.
- [writing] Figure reference formatting inconsistency: In Section 'e001', figure references use mixed casing (e.g., 'Figures~\ref{Fig:ADD}' vs 'Figure~\ref{User_Study}(a)'). Ensure consistent capitalization ('Figure' vs 'Fig.') and spacing before the reference command throughout the document.
- [writing] LaTeX hygiene in tables: Several supplementary tables (e.g., 'table/Algorithm_Visual_Reason.tex') use '\resizebox' which can distort font sizes and lead to inconsistent line heights compared to non-resized tables. Consider using '\small' or '\footnote-size' with '\begin{tabular}' adjustments instead of '\resizebox' for better typographic quality.
- [writing] Citation style inconsistency: The bibliography uses mixed citation commands (\cite, \citep, \citet) without a clear pattern. For example, 'Nano-Banana Pro' is cited as \cite{nanobananapro} in text but \citep{nanobananapro} in tables. Standardize to one command (likely \cite or \citep) based on the document class requirements.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript presents a comprehensive benchmark with a generally clear structure, but several mechanical and stylistic inconsistencies detract from the overall polish.

First, there is a notable inconsistency in the naming of the proprietary model "Nano-Banana Pro." The Abstract and Table 1 refer to it as "Nano Banana Pro" (no hyphen), while the Introduction and Section 5.2 use "Nano-Banana Pro" (with a hyphen). The authors should standardize this nomenclature throughout the text to maintain professional consistency.

Second, a typographical error appears in the table header in Section 5.2 (Table 3), where "Multimodel" is used instead of the correct term "Multimodal" in the phrase "Open-source Multimodel Large Language Models." This should be corrected to align with standard terminology used elsewhere in the paper.

Third, in the Appendix, the prompt box labeled 'IF_Multi-Image' contains a title "Instruction Following System Prompts for **Mutli**-Image Tasks." The word "Mutli" is a typo and must be corrected to "Multi."

Finally, the narrative in Section 5.2 regarding the performance of Qwen3.5-9B requires slight clarification. The text states it "rivals larger models," yet Table 3 shows Qwen3.6-27B (a larger model) achieving a higher average score (0.7183 vs 0.6681). While the claim may hold in specific sub-metrics, the phrasing is slightly ambiguous given the aggregate data. A minor rephrase to specify the context of the rivalry (e.g., "rivals larger models in specific reasoning tasks" or "achieves comparable performance to larger models with

fewer parameters”) would improve precision.

Overall, the writing is readable, but these specific edits are necessary to meet the high standards of the venue.

- [writing] In Section 1 (Introduction), the phrase ‘Nano-Banana Pro’ is inconsistently hyphenated compared to ‘Nano Banana Pro’ used in the Abstract and Table 1. Standardize the model name throughout the manuscript.
- [writing] In Section 5 (Experiments), the term ‘Multimodel’ is used in the table header for ‘Open-source Multimodel Large Language Models’. This should be corrected to ‘Multimodal’ to match standard terminology and the rest of the paper.
- [writing] In the Appendix, the label for the system prompt box reads ‘IF_Multi-Image’ but the title text says ‘Mutli-Image Tasks’. Correct the typo ‘Mutli’ to ‘Multi’ in the title text.
- [writing] In Section 5.2 (Main Results), the text states ‘Qwen3.5-9B rivals larger models’ but the table shows Qwen3.6-27B achieving the highest open-source score. Clarify the comparison to ensure the text accurately reflects the data presented in Table 3.